Practical Week-10

1.Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

```
vedadripinninti@MSI123 ~/OSLab/Week-10$ echo "Hello from vedadripinninti" > original.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ cat original.txt
Hello from vedadripinninti
vedadripinninti@MSI123 ~/OSLab/Week-10$ ls -i original.txt
13732 original.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ stat original.txt
  File: original.txt
  Size: 20            Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 13732       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/vedadripinninti)   Gid: ( 1000/vedadripinninti)
Access: 2026-09-12 10:13:14.617607567 +0000
Modify: 2026-09-12 10:12:56.021606597 +0000
Change: 2026-09-12 10:12:56.021606597 +0000
 Birth: 2026-09-12 10:12:56.021606597 +0000
vedadripinninti@MSI123 ~/OSLab/Week-10$ ln original.txt hardlink.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ ls -li original.txt hardlink.txt
13732 -rw-r--r-- 2 vedadripinninti vedadripinninti 20 Sep 12 10:12 hardlink.txt
13732 -rw-r--r-- 2 vedadripinninti vedadripinninti 20 Sep 12 10:12 original.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ echo "Modified using hard link" >> hardlink.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ cat original.txt
Hello from vedadripinninti
Modified using hard link
vedadripinninti@MSI123 ~/OSLab/Week-10$ stat original.txt
  File: original.txt
  Size: 45            Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 13732       Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/vedadripinninti)   Gid: ( 1000/vedadripinninti)
Access: 2026-09-12 10:14:34.025609030 +0000
Modify: 2026-09-12 10:14:19.201608803 +0000
Change: 2026-09-12 10:14:19.201608803 +0000
 Birth: 2026-09-12 10:12:56.021606597 +0000
vedadripinninti@MSI123 ~/OSLab/Week-10$ ln -s original.txt symlink.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ ls -li original.txt hardlink.txt symlink.txt
13732 -rw-r--r-- 2 vedadripinninti vedadripinninti 45 Sep 12 10:14 hardlink.txt
13732 -rw-r--r-- 2 vedadripinninti vedadripinninti 45 Sep 12 10:14 original.txt
13745 lrwxrwxrwx 1 vedadripinninti vedadripinninti 12 Sep 12 10:15 symlink.txt -> original.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ ls -i original.txt
13732 original.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ find . -inum 13732
./original.txt
./hardlink.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ rm original.txt
vedadripinninti@MSI123 ~/OSLab/Week-10$ cat hardlink.txt
Hello from vedadripinninti
Modified using hard link
```

2.Develop a program that redirects standard input and output using file descriptors and dup2(). Demonstrate how shells implement I/O redirection.

```
vedadripinninti@MSI123
 GNU nano 8.7.1                                          src/mmap_file.c *

#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main() {
    int fd;
    struct stat st;
    char *data;

    fd = open("mmap_input.txt", O_RDWR);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    if (fstat(fd, &st) < 0) {
        perror("fstat");
        close(fd);
        return 1;
    }

    data = mmap(NULL, st.st_size,
                PROT_READ | PROT_WRITE,
                MAP_SHARED, fd, 0);
```

```
    if (data == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    printf("Original file content:\n%s\n", data);

    strcpy(data, "Modified using mmap!");

    printf("File modified successfully.\n");

    msync(data, st.st_size, MS_SYNC);

    munmap(data, st.st_size);
    close(fd);

    return 0;
}
```

Output:

```
vedadripinninti@OSI$kill/Week-10$ echo "Hello from mmap! This file will be modified." > mmap_input.txt
vedadripinninti@OSI$kill/Week-10$ cat mmap_input.txt
Hello from mmap! This file will be modified.
vedadripinninti@OSI$kill/Week-10$ nano src/mmap_file.c

vedadripinninti@OSI$kill/Week-10$ gcc src/mmap_file.c -o bin/mmap_file
vedadripinninti@OSI$kill/Week-10$ ./bin/mmap_file
Original file content:
Hello from mmap! This file will be modified.

File modified successfully.
```